BITS Pilani

Work Integrated Learning Programmes Division

M.Tech (AIML)

Machine Learning Assignment 2

Wine Quality Classification

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1. Submission Links

1.1 GitHub Repository

URL: https://github.com/Satvik-12/ML-Assignment-2

Contains complete source code, trained model files, requirements.txt, test_data.csv, and README.md documentation.

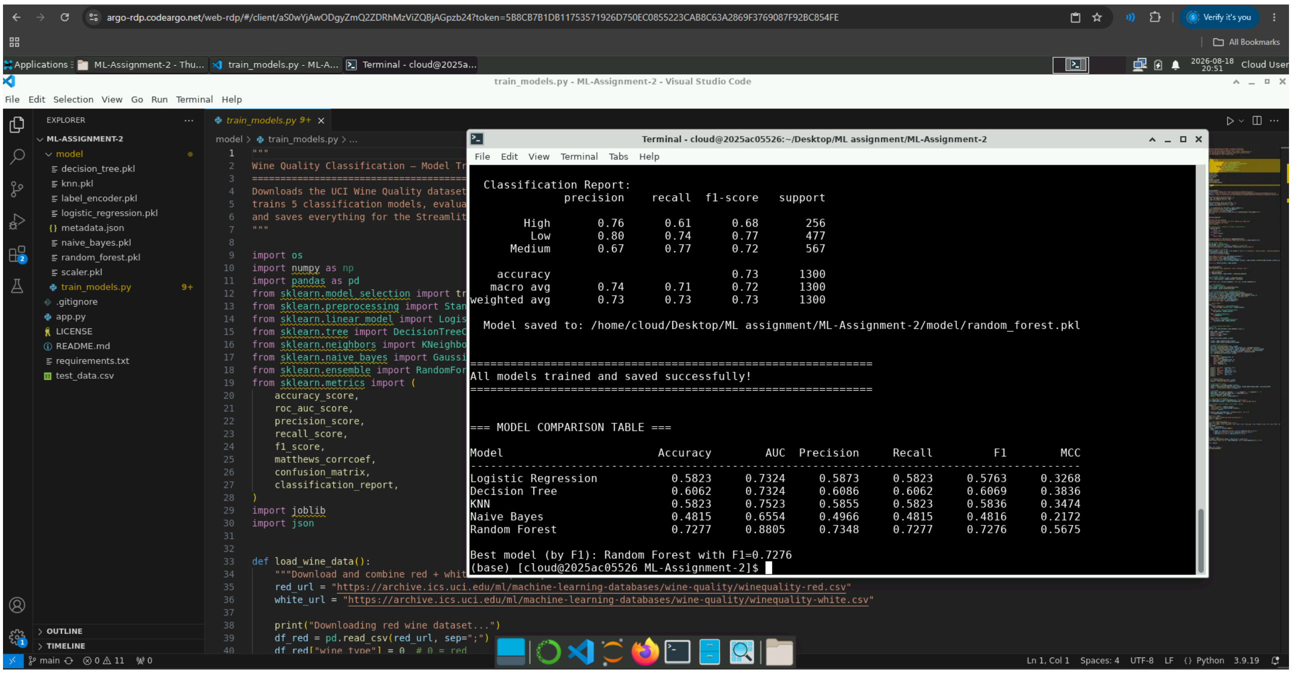
1.2 Live Streamlit App

URL: https://satvik-12-ml-assignment-2.streamlit.app/

Interactive web application deployed on Streamlit Community Cloud. Supports CSV upload, model selection, evaluation metrics display, confusion matrix visualization, and model comparison dashboard.

1.3 BITS Virtual Lab Screenshot

Screenshot of assignment execution on BITS Virtual Lab:



2. Problem Statement

The objective is to classify wine quality into three categories - Low, Medium, and High - based on 12 physicochemical properties. We implement and compare 5 machine learning classification models, evaluate them using 6 standard metrics, and deploy an interactive Streamlit web application to showcase the results.

The quality scores (originally 3-9) from the UCI Wine Quality dataset are binned into: Low (quality 3-5), Medium (quality 6), and High (quality 7-9).

3. Dataset Description

Property	Details
Dataset	UCI Wine Quality Dataset
Source	UCI ML Repository
Total Instances	6,497 (1,599 red + 4,898 white)
Number of Features	12
Target Variable	Wine Quality (3 classes)
Train/Test Split	80% / 20% (stratified)

Features Used (12)

1. Fixed Acidity
2. Volatile Acidity
3. Citric Acid
4. Residual Sugar
5. Chlorides
6. Free Sulfur Dioxide
7. Total Sulfur Dioxide
8. Density
9. pH
10. Sulphates
11. Alcohol
12. Wine Type (0=Red, 1=White)

4. Models & Evaluation Metrics

4.1 Comparison Table - All Models

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.5823	0.7324	0.5873	0.5823	0.5763	0.3268
Decision Tree	0.6062	0.7324	0.6086	0.6062	0.6069	0.3836
kNN	0.5823	0.7523	0.5855	0.5823	0.5836	0.3474
Naive Bayes	0.4815	0.6554	0.4966	0.4815	0.4816	0.2172
Random Forest (Ensemble)	**0.7277**	**0.8805**	**0.7348**	**0.7277**	**0.7276**	**0.5675**

** Green row = Best performing model (by F1 Score)*

4.2 Observations on Model Performance

ML Model Name	Observation about model performance
Logistic Regression	Achieves moderate accuracy (58.23%). Being a linear model, it struggles with non-linear feature interactions in wine quality data. The weighted AUC (0.7324) suggests reasonable class separability, but the lower MCC (0.3268) indicates limited ability to handle the multi-class imbalance effectively. Precision and recall are closely matched, indicating no strong prediction bias.
Decision Tree	Slightly better than Logistic Regression with 60.62% accuracy. The tree structure captures some non-linear relationships, improving recall across all classes. However, it shares the same AUC (0.7324) with Logistic Regression, suggesting similar probabilistic discrimination. The MCC of 0.3836 shows moderate balanced performance. Prone to overfitting despite max_depth constraint.
kNN	Matches Logistic Regression in accuracy (58.23%) but achieves a slightly higher AUC (0.7523), suggesting better probability calibration. KNN is sensitive to feature scaling (handled via StandardScaler) and the curse of dimensionality with 12 features. Performance is limited by the local neighborhood approach, which may not capture global patterns.
Naive Bayes	Lowest performing model with 48.15% accuracy. The conditional independence assumption is strongly violated by the correlated physicochemical properties (e.g., density correlates with alcohol and residual sugar). The AUC (0.6554) and MCC (0.2172) are the weakest, confirming that Gaussian NB is a poor fit for this dataset's feature distributions.
Random Forest (Ensemble)	Best performing model across all 6 metrics. Accuracy of 72.77%, AUC of 0.8805, and MCC of 0.5675 demonstrate strong classification ability. The ensemble of 100 decision trees effectively captures complex non-linear interactions between features while averaging out individual tree variance. High AUC indicates excellent class separability.

Overall Winner	Random Forest (Ensemble) - Outperforms all other models by a significant margin on every metric. The ensemble approach is particularly well-suited for this dataset due to the complex, non-linear relationships between physicochemical properties and wine quality.

5. GitHub Repository & Links

5.1 Repository Link

GitHub: https://github.com/Satvik-12/ML-Assignment-2

5.2 Streamlit App Link

App URL: https://satvik-12-ml-assignment-2.streamlit.app/

5.3 Repository Structure

```
ML-Assignment-2/
|-- app.py                    # Streamlit web application
|-- requirements.txt          # Python dependencies
|-- README.md                 # Full documentation
|-- test_data.csv             # Test dataset (1,300 samples)
|-- model/
    |-- train_models.py       # Training & evaluation script
    |-- logistic_regression.pkl
    |-- decision_tree.pkl
    |-- knn.pkl
    |-- naive_bayes.pkl
    |-- random_forest.pkl
    |-- scaler.pkl            # StandardScaler
    |-- label_encoder.pkl     # LabelEncoder
    |-- metadata.json         # Results & metadata
```

6. Streamlit App Features

a. Dataset Upload (CSV)

File uploader in sidebar - supports uploading test data CSV files. The app also auto-loads the built-in test dataset when no file is uploaded.

b. Model Selection Dropdown

Dropdown menu to choose from all 5 trained models: Logistic Regression, Decision Tree, KNN, Naive Bayes, and Random Forest.

c. Evaluation Metrics Display

Six metrics displayed as styled cards: Accuracy, AUC, Precision, Recall, F1 Score, and MCC. Also includes a full comparison table across all models with best/worst highlighting.

d. Confusion Matrix

Seaborn heatmap visualization showing the confusion matrix for the selected model, with annotated counts and clear axis labels.